

Proactive Diet Shopping Assistant

A smart shopping assistant built with Expo + React Native. It identifies food products via barcode scanning and provides real-time risk alerts and alternative suggestions based on the user's dietary goals, allergens, and family member needs.

Core Features

- Barcode scanning (EAN/UPC) — requires merchant partnership for full coverage
- Product nutrition information parsing (per 100g)
- Allergen conflict detection (including family member dimension)
- Shopping cart nutrition summary and compliance scoring
- Product comparison and alternative recommendations

Product Identification

Barcode Scanning

The app queries [Open Food Facts](#) for product data. To maximize recognition coverage in Hong Kong and Mainland China, the following strategies are applied:

1. **Region-aware query routing:** barcode prefixes are used to route queries to the appropriate Open Food Facts instance
 - Mainland China barcodes (690–699) → priority to `cn.openfoodfacts.org`
 - Other regions (including Hong Kong 489) → priority to `world.openfoodfacts.org`
2. Concurrent queries with timeout control for faster first-response
3. Automatic search fallback when direct lookup fails
4. In-memory caching with in-flight request deduplication

Note: Barcode-based product identification relies on data availability in Open Food Facts. Full coverage requires collaboration with merchants and brands to populate the database.

Camera-Based Recognition (VLM Match)

For products without a scannable barcode, the app supports camera-based recognition via **VLM Match (Vision-Language Model Matching)**. This approach uses a vision-language model to analyze product images and match them against known product information, enabling identification of products that are not yet in any database.

Current Status: Camera-based VLM Match is under development and not yet fully functional.

Tech Stack

- Expo 54
- React 19
- React Native 0.79
- Expo Router

- Zustand + AsyncStorage
- React Native Paper
- i18next

Project Structure

```
1 | src/  
2 |   app/                # Route entry points (expo-router)  
3 |   screens/            # Pages  
4 |   components/         # UI components  
5 |   services/           # API, barcode logic, nutrition analysis  
6 |   stores/             # Zustand state management  
7 |   utils/              # Business rules and calculation utilities  
8 |   constants/          # Theme, goals, allergen constants  
9 |   i18n/               # Internationalization
```

Getting Started

1) Install dependencies

```
1 | npm install
```

2) Start the development server

```
1 | npx expo start
```

Shortcut commands:

```
1 | npm run android  
2 | npm run ios  
3 | npm run web
```

Code Quality

Run lint:

```
1 | npm run lint
```

Known Limitations

- Open Food Facts data completeness varies by region and brand coverage
- Camera-based VLM Match recognition is not yet fully implemented
- Offline mode does not support remote product identification (local cache only)

License

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